CHMT-Q: Causal Hierarchical Market Transformer with Quantiles

Dhruv Sharma[1]

[1]Independent Researcher, dhruv4jaipur@gmail.com

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Abstract

This paper specifies CHMT-Q, the Causal Hierarchical Market Transformer with Quantiles, from its numeric interface to its reconstructed forecasts. The model accepts four value tensors—64 one-minute bars, 240 half-second level-1 observations, 240 two-sided top-ten level-2 snapshots, and a 16-value quality vector—together with four authoritative validity masks. Separate encoders preserve the different geometry of slow bars, best quotes, and ranked depth. The level-2 branch first attends across the 20 side–rank positions of each snapshot, then pools and attends across time. Quality-aware routing joins 64, 60, and 60 encoded tokens into a 184-token memory. Five learned horizon queries decode five consecutive one-minute future intervals in parallel. The raw output has shape $[B, 5, 17]$: for each horizon, 15 center/gap parameters define ordered 10th, 50th, and 90th percentiles for five footprint targets, and two logits describe buy- and sell-point-of-control presence.

We give the complete feature and mask contract, every tensor shape, physical time encoding, Q/K-normalized attention, safe all-masked softmax, hierarchical depth pooling, quality gates, decoder equations, output indexing, inverse target transformations, and the composite training objective. We also reconcile the exact 2,089,554 trainable parameters from primitive layer formulas. Here "causal" means temporal availability and masking, not causal inference, and "Q" means quantiles, not Q-learning. CHMT-Q is a forecast provider rather than an order policy. This architecture report makes no live-deployment, economic-performance, or profitability claim.

Keywords: *Transformer*, *limit order book*, *market microstructure*, *quantile regression*, *multi-horizon forecasting*, *numeric time series*

1 Scope and Contributions

CHMT-Q is a compact numeric Transformer for multi-horizon forecasting from heterogeneous electronic-market state. Its design starts from a structural observation: a minute bar is one temporal aggregate, a level-1 row is one best-quote state, and a level-2 row is a ranked two-sided set observed at one instant. Treating all three as interchangeable tokens would discard side/rank geometry or inflate the sequence with thousands of depth-level tokens.

The architecture therefore uses three independent encoders and a direct horizon-query decoder. It combines ideas from attention [1, 2], patch-based time-series models [5], multi-horizon forecasting [6], and structured limit-order-book learning [7–11], while fixing a model-specific interface and masking discipline.

This paper contributes five items:

1. an exact external contract for four value tensors and four masks;
2. an input-to-output derivation that preserves bar, quote, side, rank, and elapsed-time semantics;
3. a fully specified CHMT block built from primitive operations, including defined behavior when every key is masked;
4. an explicit map from the 17 raw numbers at each horizon to ordered quantiles, presence probabilities, and physical units; and
5. a literal parameter reconciliation totaling 2,089,554 trainable scalars.

The report is deliberately about mechanism. The output targets are future footprint descriptors. They are not OHLC candles, executable returns, trading positions, or orders. Any policy, execution model, or capital-allocation rule is a separate system with a separate validation obligation.

2 Causal Forecasting Problem

2.1 Decision time and information boundary

Let t_c be the wall-clock `FeatureCutoffUtc`: the close of the latest feature interval and the open of the first forecast interval. Each observation x_i is assigned an event time and compiler-defined availability time; a distinct local receive time is used only when captured. The archive used for the existing offline study did not contain an independent local receive timestamp.

The compiler orders records deterministically by availability time, event time, feed epoch, stable stream rank, native sequence, and payload identity. Insert a logical cutoff marker $c = t_c^-$ into that total order immediately after records assigned to the predecessor feature phase and immediately before records assigned to the target-open phase. Writing κ_i for record i's total-order key, the legal information set is

$$\mathcal{F}_c = \sigma(x_i : \kappa_i \preceq c)\,. \tag{2.1}$$

This is an archive-relative causality contract: it uses captured timestamps, source epochs, sequence identity, and deterministic event order. It neither asserts causal influence nor independently verifies physical wire-arrival time beyond the recorded capture provenance. The model is a deterministic function of that information and fixed model state:

$$\widehat{Y}_{c,1:H} = f_\theta(\mathcal{F}_c, M_c), \qquad H = 5, \tag{2.2}$$

where M_c contains immutable configuration and normalization constants, not future or cross-sample state. The five targets occupy consecutive half-open one-minute intervals on the versioned exchange grid:

$$I_h = [t_c + (h-1)\Delta, t_c + h\Delta), \qquad h \in \{1, \dots, 5\}, \quad \Delta = 60 \text{ s}. \tag{2.3}$$

Thus horizon five means the fifth future minute, not one cumulative five-minute label. In the total order, $c \prec t_c = \min_h \inf I_h$, while physical ages are measured relative to wall-clock t_c. The deterministic tie rule assigns every same-timestamp record to exactly one side of the marker, so no record can be both a feature and a label for the same example.

2.2 Five targets per interval

For interval I_h, let P_h, P_h^{buy}, and P_h^{sell} be total, buy, and sell point of control (POC); let V_h be total executed volume; and let D_h be classified buy volume minus classified sell volume. With causal reference price p_{ref}, tick size δ_p, and trailing log-volume reference ℓ_{ref}, the training targets are

$$\begin{aligned}
y_{h,0} &= \frac{P_h - p_{\text{ref}}}{\delta_p}, & y_{h,1} &= \frac{P_h^{\text{buy}} - p_{\text{ref}}}{\delta_p},\\
y_{h,2} &= \frac{P_h^{\text{sell}} - p_{\text{ref}}}{\delta_p}, & y_{h,3} &= \log(1 + V_h) - \ell_{\text{ref}},\\
y_{h,4} &= \begin{cases} \text{clip}(D_h/V_h, -1, 1)\,, & V_h > 0 \text{ and classification is valid},\\ \text{undefined}, & \text{otherwise.} \end{cases}
\end{aligned} \tag{2.4}$$

Each (h, k) has an independent target-validity mask. A buy or sell POC is undefined when the corresponding eligible classified volume is absent. An interval whose trade-tape continuity cannot be established is unknown rather than a zero-volume observation. In every undefined branch, $m^Y_{h,k} = 0$, and the stored numeric placeholder is excluded from all loss terms.

For continuous targets, frozen per-horizon, per-target statistics fitted only on the training partition define

$$\widetilde{y}_{h,k} = \frac{y_{h,k} - \mu^Y_{h,k}}{s^Y_{h,k}}, \qquad s^Y_{h,k} > 0. \tag{2.5}$$

The bounded delta fraction uses the identity transform before the output head's monotone tanh map. Equation (2.5) is reversed before any physical-unit reconstruction.

3 Complete External Input Contract

3.1 Value tensors and authoritative masks

For training batch size B, the four value tensors are

$$\begin{aligned} X^b &\in \mathbb{R}^{B\times 64\times 16}, \qquad X^1 \in \mathbb{R}^{B\times 240\times 16}, \\ X^2 &\in \mathbb{R}^{B\times 240\times 2\times 10\times 12}, \qquad q \in \mathbb{R}^{B\times 16}. \end{aligned} \tag{3.1}$$

They are accompanied by

$$\begin{aligned} m^b &\in \{0,1\}^{B\times 64}, \qquad m^1 \in \{0,1\}^{B\times 240}, \\ m^{2,L} &\in \{0,1\}^{B\times 240\times 2\times 10}, \qquad m^{2,S} \in \{0,1\}^{B\times 240}. \end{aligned} \tag{3.2}$$

The exported inference graph fixes $B = 1$ without changing any semantic axis. Invalid rows are numerically zero after transformation, but zero is a possible valid value. Equations (3.2), not the numeric contents, decide eligibility.

Name	Training shape	Dtype	Meaning
`bars`	$[B, 64, 16]$	float32	64 consecutive finalized one-minute clock slots, oldest first
`bars_mask`	$[B, 64]$	bool	legal and available bar slots
`l1`	$[B, 240, 16]$	float32	120 seconds on a 500-ms best-quote grid
`l1_mask`	$[B, 240]$	bool	accepted, available, and fresh L1 rows
`l2`	$[B, 240, 2, 10, 12]$	float32	120 seconds of bid/ask top-ten depth
`l2_level_mask`	$[B, 240, 2, 10]$	bool	eligible side–rank cells
`l2_snapshot_mask`	$[B, 240]$	bool	accepted snapshots containing an eligible level
`quality`	$[B, 16]$	float32	causal coverage, carry, gap, and anchor facts

Table 1: External model inputs. No text, symbol identifier, tokenizer output, or arbitrary categorical ID crosses the model boundary.

Training additionally supplies targets $Y \in \mathbb{R}^{B\times 5\times 5}$, target mask $m^Y \in \{0,1\}^{B\times 5\times 5}$, side-presence labels $Z \in \{0,1\}^{B\times 5\times 2}$, and presence mask $m^P \in \{0,1\}^{B\times 5\times 2}$. These tensors supervise the model but are not inference inputs.

The level and snapshot masks obey the invariant

$$m^{2,S}_{b,t} = a^2_{b,t} \wedge \bigvee_{s=0}^{1} \bigvee_{r=0}^{9} m^{2,L}_{b,t,s,r}, \tag{3.3}$$

where a^2 denotes accepted-and-fresh snapshot status. A one-sided book may retain valid cells on one side and mask the other. A true snapshot mask with no valid level is rejected.

3.2 Bar feature vector

The bar axis is the 64 consecutive clock slots ending at wall-clock t_c; it is not the last 64 nonempty bars. Let p_- be the previous valid close and g the preceding missing-slot count.

Table 2: The 16 bar fields.

Index	Field	Causal definition
0–3	OHLC offsets	Each price minus p_-, divided by δ_p
4	log return	$\log(p_{\text{close}}/p_-)$ for positive prices
5	range ticks	$(p_{\text{high}} - p_{\text{low}})/\delta_p$
6	body ticks	$(p_{\text{close}} - p_{\text{open}})/\delta_p$
7	upper-wick ticks	$(p_{\text{high}} - \max(p_{\text{open}}, p_{\text{close}}))/\delta_p$
8	lower-wick ticks	$(\min(p_{\text{open}}, p_{\text{close}}) - p_{\text{low}})/\delta_p$
9	close location	$(p_{\text{close}} - p_{\text{low}})/\max(p_{\text{high}} - p_{\text{low}}, \delta_p)$
10	log volume	$\log(1 + V)$
11	log-volume residual	log volume minus its trailing causal exponential mean

Index	Field	Causal definition
12	availability age	$\max(0, t_c - t^{\text{available}})$ seconds
13	preceding missing slots	$\log(1 + \min(g, 63))/\log 64$
14–15	session phase	sine and cosine of versioned exchange-session progress

Only one eligible close immediately before the window may be retained to compute the oldest token's offsets and return. It is not a 65th model token. If it is unavailable, the oldest row is masked.

3.3 Level-1 feature vector

All L1 price offsets use p_{ref}. Rows are sampled as of each half-second grid time from observations already available at that time.

Table 3: The 16 L1 fields.

Index	Field	Definition
0	midpoint offset	$(p_{\text{mid}} - p_{\text{ref}})/\delta_p$
1	midpoint change	change from previous valid grid midpoint in ticks
2	spread	$(p_{\text{ask}} - p_{\text{bid}})/\delta_p$
3–4	log bid/ask size	$\log(1 + \max(v_{\text{side}}, 0))$
5	size imbalance	$(v_b - v_a)/(v_b + v_a)$ when the sum is positive
6	microprice offset	$\big(p_{\text{ask}} v_b + p_{\text{bid}} v_a\big)/(v_b + v_a) - p_{\text{mid}}$, divided by δ_p, for $v_b + v_a > 0$
7–8	bid/ask changes	price change from the previous valid grid row in ticks
9–10	bid/ask log-size changes	change from the previous valid log size
11	observation age	grid time minus availability time, seconds
12	event lag	availability time minus event time, seconds
13	sequence gap	$\min(g_{\text{quote}}, 16)/16$, where g_{quote} is the missing-sequence count
14	carried-forward flag	one when the grid row reuses an eligible earlier observation
15	approximate-time flag	one when event time is approximate

Crossed, nonpositive, stale, and unavailable books are masked and summarized in the quality vector. Forward carry is permitted only inside a fixed freshness limit; back-filling from a later observation is forbidden.

3.4 Level-2 feature vector

The L2 tensor keeps side and rank as explicit axes. Within a snapshot, each cell has the following 12 numeric fields.

Table 4: The 12 L2 side–rank fields.

Index	Field	Definition
0	signed distance	$(p_{s,r} - p_{\text{mid}})/\delta_p$ using a fresh grid-time midpoint
1	better-level gap	$\lvert p_{s,r} - p_{s,r-1}\rvert/\delta_p$ for $r > 0$; zero at rank zero
2	log size	$\log(1 + \max(v_{s,r}, 0))$
3	log cumulative size	log of one plus same-side cumulative size through rank r
4	side-size share	level size divided by total eligible top-ten same-side size
5	cumulative-side share	cumulative size divided by total eligible same-side size
6	paired-rank imbalance	bid/ask size imbalance at the same rank
7	snapshot age	grid time minus availability time, seconds
8	event lag	availability time minus event time, seconds
9	sequence gap	$\min(g_{\text{depth}}, 16)/16$, where g_{depth} is the missing-sequence count
10	carried-forward flag	one when the row reuses an eligible earlier snapshot
11	approximate-time flag	one when event time is approximate

Bid/ask side $s \in \{0, 1\}$ and rank $r \in \{0, \ldots, 9\}$ are fixed structural axes. Learned internal side and rank embeddings represent them; the market-data pipeline does not pass token IDs.

3.5 Quality vector

The quality tensor records causal facts about exactly the compiled windows.

Table 5: The 16 quality fields.

Index	Field	Index	Field
0	valid-bar fraction	8	quote-sequence-gap rate
1	valid-L1 fraction	9	depth-sequence-gap rate
2	valid-L2-snapshot fraction	10	late-bar-availability fraction
3	valid-L2-level fraction	11	out-of-order-observation rate
4	L1 carried fraction	12	one-sided-book rate
5	L2 carried fraction	13	crossed/rejected-book rate
6	approximate-L1-time fraction	14	anchor-from-L1 flag
7	approximate-L2-time fraction	15	normalized session progress

Rates use only information eligible by logical cutoff c. An empty rate denominator yields zero together with a separate hard-availability fact; it does not fabricate an observation.

4 End-to-End Tensor Flow

Stage	Bars	L1	L2	Quality / decoder
External	$[B, 64, 16]$	$[B, 240, 16]$	$[B, 240, 2, 10, 12]$	$[B, 16]$
Projection	$[B, 64, 128]$	$[B, 240, 128]$	$[B \cdot 240, 20, 32]$	$[B, 128]$
Structure	unchanged	unchanged	$[B, 240, 128]$	gates $[B, 3]$
Patching	**none**	$[B, 60, 128]$	$[B, 60, 128]$	z_{ctx} $[B, 128]$
Temporal encoder	$[B, 64, 128]$	$[B, 60, 128]$	$[B, 60, 128]$	memory $[B, 184, 128]$
Direct decoder	—	—	—	queries $[B, 5, 128]$
Raw head	—	—	—	$[B, 5, 17]$
Semantic output	—	—	—	quantiles $[B, 5, 5, 3]$; presence $[B, 5, 2]$

Table 6: Exact shape trace. The three encoders do not share weights.

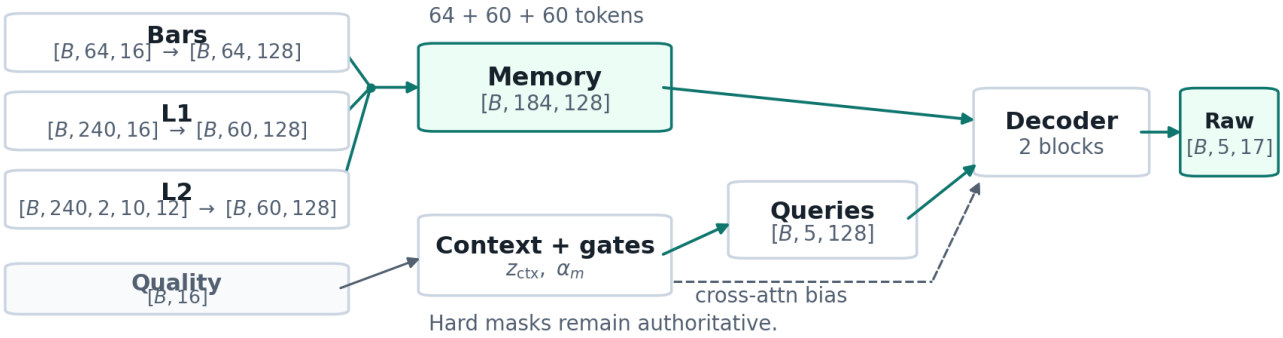


Figure 1: The three branches converge only after structural and temporal encoding. Masks remain parallel to all value tensors.

For one inference request, $B = 1$. The main semantic path is therefore

$$\begin{aligned}&[1, 64, 16],\ [1, 240, 16],\ [1, 240, 2, 10, 12],\ [1, 16]\\ &\quad\longrightarrow [1, 64, 128],\ [1, 60, 128],\ [1, 60, 128]\\ &\quad\longrightarrow [1, 184, 128] \longrightarrow [1, 5, 128] \longrightarrow [1, 5, 17].\end{aligned} \tag{4.1}$$

5 Normalization, Time, and Structural Embeddings

5.1 Training-only robust scaling

For each continuous feature j, valid training-partition values determine

$$\mu_j = \mathrm{median}_{\mathrm{train}}(x_j), \qquad s_j = \max(1.4826\,\mathrm{MAD}_{\mathrm{train}}(x_j), s_{\min,j})\,, \tag{5.1}$$

and the deployed transform is

$$z_j = \mathrm{clip}\left(\frac{x_j - \mu_j}{s_j}, -8, 8\right). \tag{5.2}$$

Bounded shares, fractions, imbalances, and binary flags use identity scaling. Invalid values are zeroed only after Equation (5.2); masks are never scaled.

At a 500-ms grid time g_t, the eligible source observation is the latest available observation not later than the grid time:

$$i^*(t) = \arg\max_i \{t_i^{\mathrm{available}} : t_i^{\mathrm{available}} \le g_t\}, \tag{5.3}$$

subject to feed-epoch continuity, acceptance, and freshness. This as-of rule preserves a fixed clock axis without future fill.

5.2 Physical-time encoding

Text position is not an adequate description of irregular market age. For nonnegative age a, context maximum $a_{\max}$, and fixed scales $\tau_1, \ldots, \tau_K$, define

$$\phi(a) = \left[\frac{a}{a_{\max}}, \frac{\log(1+a)}{\log(1+a_{\max})}, \sin\frac{a}{\tau_1}, \cos\frac{a}{\tau_1}, \ldots, \sin\frac{a}{\tau_K}, \cos\frac{a}{\tau_K}\right]. \tag{5.4}$$

The branch-specific time embedding is

$$e_t(a) = W_{t,2}\,\mathrm{SiLU}(W_{t,1}\phi(a) + b_{t,1}) + b_{t,2}. \tag{5.5}$$

Bars use $K = 7$ and scales $(60, 120, 240, 480, 960, 1920, 3840)$ seconds. Patched L1 and L2 use $K = 8$ and scales $(0.5, 1, 2, 4, 8, 16, 32, 64)$ seconds. Bar slot-close ages are $(3780, 3720, \ldots, 0)$ seconds; patch-end ages are $(118, 116, \ldots, 0)$ seconds. These arrays are constants, not external inputs.

Every temporal token also receives a learned modality vector. An L2 level state receives fixed-axis side and rank embeddings:

$$x^{2,(0)}_{b,t,s,r} = W_{\mathrm{level}} z_{b,t,s,r} + b_{\mathrm{level}} + e^{\mathrm{side}}_s + e^{\mathrm{rank}}_r. \tag{5.6}$$

5.3 Learned relative-time bias

Temporal attention head j gives an eligible earlier key a learned recency bias

$$\beta^{(j)}_{i\ell} = -\,\mathrm{softplus}(\gamma_j)\log\left(1 + \frac{t_i - t_\ell}{s_m}\right), \qquad t_\ell \le t_i, \tag{5.7}$$

where $s_m = 60$ seconds for bars and $s_m = 2$ seconds for patched microstructure. This is a soft content-independent prior. It does not replace the causal or validity mask.

6 The CHMT Primitive Block

6.1 RMS normalization

For $x \in \mathbb{R}^d$, learned gain $g \in \mathbb{R}^d$, and $\epsilon > 0$,

$$\mathrm{RMSNorm}(x) = g \odot \frac{x}{\sqrt{d^{-1}\sum_{r=1}^{d} x_r^2 + \epsilon}}. \tag{6.1}$$

RMSNorm controls scale without subtracting a coordinate mean [3].

6.2 Q/K-normalized multi-head attention

Let query states $U \in \mathbb{R}^{B\times n_q\times d}$ and memory states $M \in \mathbb{R}^{B\times n_k\times d}$. For head j with width $d_h = d/H_a$,

$$Q_j = UW_j^Q + b_j^Q, \quad K_j = MW_j^K + b_j^K, \quad V_j = MW_j^V + b_j^V. \tag{6.2}$$

Each query and key vector is RMS-normalized without a learned gain:

$$\bar{q} = \frac{q}{\sqrt{d_h^{-1}\|q\|_2^2 + \epsilon}}, \qquad \bar{k} = \frac{k}{\sqrt{d_h^{-1}\|k\|_2^2 + \epsilon}}. \tag{6.3}$$

With learned log-temperature θ_j, optional time bias β, and eligibility mask $\mathcal{M}$, define the cross-attention key-modality bias as $g_{m(\ell)} = \log(\alpha_{m(\ell)} + \epsilon)$; it is zero in attention layers that do not use modality routing. The score is

$$s_{i\ell}^{(j)} = e^{\theta_j}\frac{\bar{q}_i^{(j)\top}\bar{k}_\ell^{(j)}}{\sqrt{d_h}} + \beta_{i\ell}^{(j)} + g_{m(\ell)} + \mathcal{M}_{i\ell}. \tag{6.4}$$

In temporal self-attention, $\mathcal{M}_{i\ell}$ permits a key only when it is valid and no later than the query. In L2 within-snapshot attention there is no causal triangle because all 20 positions coexist at one logical time. Decoder cross-attention sees every valid historical memory key.

6.3 Safe masked softmax

Ordinary masked softmax is undefined when a query has no eligible key. Let $m_{i\ell} \in \{0,1\}$ and $a_i = \bigvee_\ell m_{i\ell}$. Totalize the row maximum as

$$s_i^{\max} = \begin{cases} \max_{\ell: m_{i\ell}=1} s_{i\ell}, & a_i = 1, \\ 0, & a_i = 0, \end{cases} \tag{6.5}$$

and define

$$r_{i\ell} = m_{i\ell}\exp(s_{i\ell} - s_i^{\max}), \qquad p_{i\ell} = \begin{cases} r_{i\ell}/\sum_k r_{ik}, & a_i = 1, \\ 0, & a_i = 0. \end{cases} \tag{6.6}$$

Then every masked key has exactly zero weight, every nonempty eligible row sums to one, and every empty row returns zero without a nonfinite intermediate in the reference path. At training time, attention-weight dropout is applied after the safe softmax; at inference it is the identity. Multi-head output is

$$\mathrm{Attn}(U, M) = \mathrm{Concat}_{j=1}^{H_a}(\mathrm{Dropout}(P_j)V_j)W^O + b^O. \tag{6.7}$$

6.4 SwiGLU and residual update

The feed-forward sublayer is

$$\mathrm{FFN}(x) = W_o[\mathrm{SiLU}(W_g x + b_g) \odot (W_u x + b_u)] + b_o, \tag{6.8}$$

with main widths $128 \to 256 \to 128$ and level widths $32 \to 64 \to 32$ [4]. For encoder input x, let $\bar{x} = \mathrm{RMSNorm}_a(x)$. The pre-normalized LayerScale updates are

$$\begin{aligned} u &= x + \lambda_a \odot \mathrm{Dropout}(\mathrm{Attn}(\bar{x}, \bar{x})), \\ z &= u + \lambda_f \odot \mathrm{Dropout}(\mathrm{FFN}(\mathrm{RMSNorm}_f(u))). \end{aligned} \tag{6.9}$$

Both λ_a and λ_f are learned d-vectors initialized to 0.1. The query-validity mask multiplies z after the residual path. Dropout is 0.10 during training and disabled at inference.

7 Three Structure-Preserving Encoders

7.1 Bar encoder: 64 unpatched tokens

Each normalized bar row is independently projected from 16 to 128 dimensions, then receives its modality and physical-time embeddings:

$$X^{b,(0)} = (X^b W_b + b_b) + E_t^b + e_{\mathrm{mod}}^b. \tag{7.1}$$

Three causal CHMT blocks preserve the shape $[B, 64, 128]$. The bar branch is *not patched*: the final memory contains all 64 minute states. Its causal triangle ensures every intermediate state summarizes only its own prefix.

7.2 L1 encoder: four-frame causal patches

L1 rows are projected to width 128. Four consecutive 500-ms frames form a non-overlapping two-second patch. With projected frame x_j, validity m_j, and learned within-patch offset e_j^{patch}, the pooling score and output are

$$\begin{aligned} \widetilde{x}_j &= x_j + e_j^{\text{patch}}, \qquad j \in \{0,1,2,3\}, \\ r_j &= w_2^\top \tanh(W_1 \widetilde{x}_j + b_1) + b_2, \\ \pi_j &= \text{SafeSoftmax}_j(r_j; m_j), \qquad p = \sum_{j=0}^{3} \pi_j \widetilde{x}_j. \end{aligned} \tag{7.2}$$

An all-invalid patch returns a zero vector and false mask. Reshaping and pooling give

$$[B, 240, 128] \to [B, 60, 4, 128] \to [B, 60, 128]. \tag{7.3}$$

Patch-end time and modality embeddings are added, followed by three independent causal CHMT blocks.

7.3 L2 encoder: side/rank first, time second

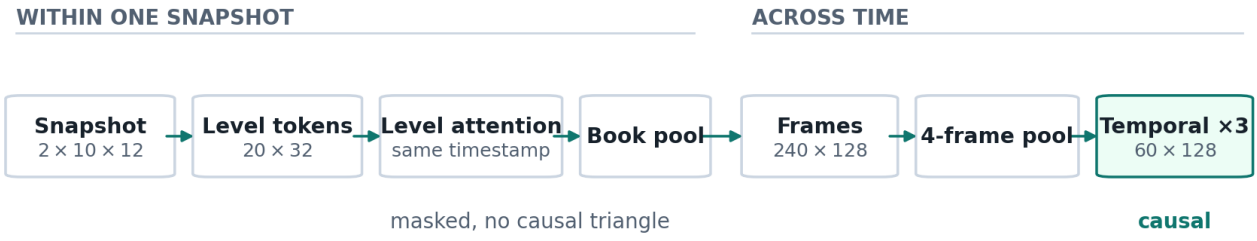


Figure 2: The L2 hierarchy separates same-time book geometry from temporal evolution. The level block is noncausal across the 20 simultaneous cells; the later temporal blocks are causal.

For every one of 240 grid times, fixed side and rank axes flatten as

$$[B, 240, 2, 10, 12] \to [B \cdot 240, 20, 12] \to [B \cdot 240, 20, 32]. \tag{7.4}$$

Equation (5.6) supplies level states. One masked, noncausal level CHMT block models interactions among the contemporaneous bid/ask ranks. For learned book query $q_{\text{book}} \in \mathbb{R}^{32}$, pool weights and the depth-frame token are

$$\begin{aligned} a_{b,t,s,r} &= \text{SafeSoftmax}_{s,r}\left(\frac{q_{\text{book}}^\top \,\text{RMSNorm}(x_{b,t,s,r})}{\sqrt{32}}; m^{2,L}_{b,t,s,r} \right), \\ d_{b,t} &= W_d \sum_{s=0}^{1} \sum_{r=0}^{9} a_{b,t,s,r} x_{b,t,s,r} + b_d \in \mathbb{R}^{128}. \end{aligned} \tag{7.5}$$

This produces $[B, 240, 128]$. The independently parameterized four-frame pool reduces it to $[B, 60, 128]$, and three causal temporal CHMT blocks encode its two-minute evolution.

The hierarchy avoids a flat 4,800-token temporal sequence. A flat attention matrix would contain $4{,}800^2 = 23{,}040{,}000$ scores per head and layer, while also mixing same-snapshot rank structure with temporal order.

8 Quality-Aware Fusion

Let $M_b \in \mathbb{R}^{B\times 64\times 128}$, $M_1 \in \mathbb{R}^{B\times 60\times 128}$, and $M_2 \in \mathbb{R}^{B\times 60\times 128}$ denote encoded memories and let m_m denote their token masks. The latest valid index and summary are

$$t_m^*(b) = \max\{t : m_{m,b,t} = 1\}, \qquad s_{m,b} = M_{m,b,t_m^*(b)}, \tag{8.1}$$

with a zero summary and hard availability $a_{m,b} = 0$ when the set is empty. The quality MLP is

$$c_q = W_{q,2}\,\text{SiLU}(W_{q,1} q + b_{q,1}) + b_{q,2} \in \mathbb{R}^{B\times 128}. \tag{8.2}$$

Each modality has an independent gate:

$$\widetilde{g}_m = a_m \,\mathrm{sigmoid}\big(w_{m,2}^\top \,\mathrm{SiLU}(W_{m,1}[s_m; c_q] + b_{m,1}) + b_{m,2}\big)\,,$$
$$\alpha_m = \frac{\widetilde{g}_m}{\sum_n \widetilde{g}_n + \epsilon}, \qquad z_{\mathrm{ctx}} = \mathrm{RMSNorm}\left(c_q + \sum_m \alpha_m s_m\right). \tag{8.3}$$

The historical memory is a concatenation, not a pooled summary:

$$M = [M_b; M_1; M_2] \in \mathbb{R}^{B\times 184\times 128}. \tag{8.4}$$

Decoder cross-attention receives $\log(\alpha_m + \epsilon)$ as the score bias for keys from modality m. The gate does not also multiply memory values. Hard-masked branches remain ineligible regardless of learned gate values. If all three modalities are unavailable, the host abstains before model execution.

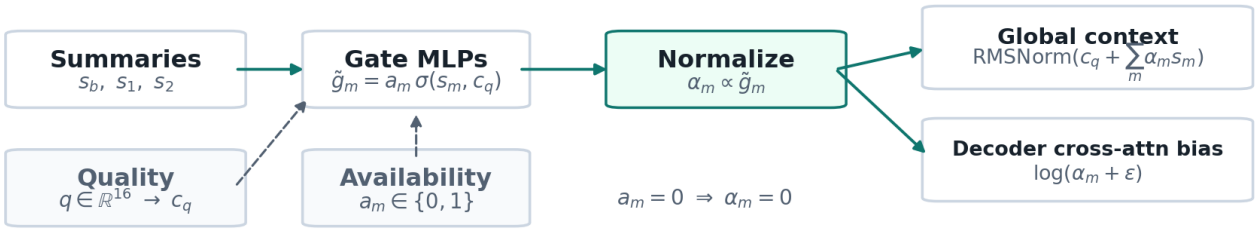


Figure 3: Hard availability bounds the learned reliability gates. A soft gate may downweight available evidence but cannot restore a masked branch.

9 Direct Five-Horizon Decoder

For horizon $h \in \{1, \ldots, 5\}$, let $e_h \in \mathbb{R}^{128}$ be a learned base query. With shared horizon MLP output u_h and context projection W_{ctx}, initialize

$$\xi_h = [h/5,\ \log(1+h),\ h\Delta],$$
$$u_h = W_{h,2}\,\mathrm{SiLU}(W_{h,1}\xi_h + b_{h,1}) + b_{h,2},$$
$$Q_h^{(0)} = e_h + W_{\mathrm{ctx}} z_{\mathrm{ctx}} + b_{\mathrm{ctx}} + u_h. \tag{9.1}$$

Stacking gives $Q^{(0)} \in \mathbb{R}^{B\times 5\times 128}$. Decoder block ℓ performs three residual updates in a fixed order:

$$A^{(\ell)} = Q^{(\ell)} + \lambda_s^{(\ell)} \odot \mathrm{Attn}_s(\mathrm{RMSNorm}(Q^{(\ell)}), \mathrm{RMSNorm}(Q^{(\ell)})),$$
$$C^{(\ell)} = A^{(\ell)} + \lambda_c^{(\ell)} \odot \mathrm{Attn}_c(\mathrm{RMSNorm}(A^{(\ell)}), M),$$
$$Q^{(\ell+1)} = C^{(\ell)} + \lambda_f^{(\ell)} \odot \mathrm{FFN}(\mathrm{RMSNorm}(C^{(\ell)})), \qquad \ell \in \{0, 1\}. \tag{9.2}$$

Dropout surrounds each residual branch during training and is omitted from Equation (9.2) for readability. Horizon self-attention is noncausal because all five queries are constructed at the same decision time. No predicted horizon is used as input to another; all five forecasts are direct functions of the same historical memory. Each decoder block has exactly three learned RMSNorm gains—one before query self-attention, one before the cross-attention query, and one before the feed-forward layer. No additional learned RMSNorm is applied to memory M inside cross-attention.

10 Raw Output, Ordered Quantiles, and Physical Units

10.1 Exact 17-slot map

The shared head is

$$R = Q^{(2)} W_R + b_R \in \mathbb{R}^{B\times 5\times 17}. \tag{10.1}$$

Slots	Target	Raw meaning	Semantic meaning after transforms
0–2	total POC	center, lower-gap logit, upper-gap logit	q10/q50/q90 tick offset
3–5	buy POC	center, lower-gap logit, upper-gap logit	conditional q10/q50/q90 tick offset
6–8	sell POC	center, lower-gap logit, upper-gap logit	conditional q10/q50/q90 tick offset
9–11	log-volume residual	center, lower-gap logit, upper-gap logit	q10/q50/q90 future volume
12–14	delta fraction	center, lower-gap logit, upper-gap logit	bounded q10/q50/q90 fraction
15	buy POC presence	logit	probability that an eligible buy POC exists
16	sell POC presence	logit	probability that an eligible sell POC exists

Table 7: Target-major raw-head indexing for each horizon.

It emits raw parameters, not already reconstructed quantiles. For target $k \in \{0, \dots, 4\}$,

$$(r_{h,k}, a_{h,k}, b_{h,k}) = (R_{h,3k}, R_{h,3k+1}, R_{h,3k+2}). \tag{10.2}$$

The two presence probabilities are

$$\widehat{p}_h^{\text{buy}} = \text{sigmoid}(R_{h,15}), \qquad \widehat{p}_h^{\text{sell}} = \text{sigmoid}(R_{h,16}). \tag{10.3}$$

They indicate target existence under the label contract. They do not indicate trade direction.

10.2 Noncrossing quantile construction

For every (h, k), define normalized quantiles

$$\widetilde{Q}_{h,k,0.50} = r_{h,k}, \qquad \widetilde{Q}_{h,k,0.10} = r_{h,k} - \text{softplus}(a_{h,k}), \qquad \widetilde{Q}_{h,k,0.90} = r_{h,k} + \text{softplus}(b_{h,k}). \tag{10.4}$$

The delta-fraction target uses its bounded semantic value inside the loss. Define the post-head prediction

$$Q^{\star}_{h,k,\tau} = \begin{cases} \widetilde{Q}_{h,k,\tau}, & k \in \{0,1,2,3\}, \\ \tanh(\widetilde{Q}_{h,4,\tau}), & k = 4. \end{cases} \tag{10.5}$$

Proposition 1 (Order preservation)**.** *Equation* (10.4) *satisfies* $\widetilde{Q}_{h,k,0.10} < \widetilde{Q}_{h,k,0.50} < \widetilde{Q}_{h,k,0.90}$ *over exact real arithmetic. Any nondecreasing inverse transformation preserves weak order.*

Proof. $\text{softplus}(x) = \log(1 + e^x) > 0$ for every finite x. Therefore $r - \text{softplus}(a) < r < r + \text{softplus}(b)$. If f is nondecreasing and $x_1 < x_2 < x_3$, then $f(x_1) \le f(x_2) \le f(x_3)$. This applies to affine inverse scaling, the exponential volume map, and tanh for delta fraction. Finite-precision saturation may turn a strict inequality into equality but cannot reverse it. □

10.3 Inverse normalization and reconstruction

For price and log-volume targets, first reverse Equation (2.5) independently for every (h, k, τ):

$$\widehat{y}_{h,k,\tau} = \mu^Y_{h,k} + s^Y_{h,k} Q^{\star}_{h,k,\tau}, \qquad k \in \{0,1,2,3\}. \tag{10.6}$$

The primitive physical outputs are then

$$\begin{aligned} \widehat{P}_{h,k,\tau} &= p_{\text{ref}} + \delta_p \widehat{y}_{h,k,\tau}, && k \in \{0,1,2\}, \\ \widehat{V}_{h,\tau} &= \max(0, \exp(\ell_{\text{ref}} + \widehat{y}_{h,3,\tau}) - 1), \\ \widehat{f}_{D,h,\tau} &= Q^{\star}_{h,4,\tau}. \end{aligned} \tag{10.7}$$

Prices are snapped to the instrument tick only after reconstruction. Finite values, quantile order, nonnegative volume, bounded delta fraction, presence decisions, and the five-horizon shape are checked before release.

The output contains three ordered marginal quantile forecasts for each of five targets at each of five horizons. It is not a full probability distribution or a coherent joint sample path. The q10–q90 pair is a nominal central 80% band; its empirical coverage is not guaranteed without separate calibration. In particular, multiplying a volume quantile by a delta-fraction quantile does not produce a valid quantile of raw delta.

10.4 Worked synthetic output-reconstruction example

Consider one syntactically valid inference request: value tensors and masks have the batch-one shapes in Table 1; the L1 and L2 windows contain 240 grid rows; the bar window contains 64 slots. Suppose the head produces, for one horizon and total-POC target,

$$(r, a, b) = (0.20, -0.50, 0.10). \tag{10.8}$$

Since $\mathrm{softplus}(-0.50) = 0.4741$ and $\mathrm{softplus}(0.10) = 0.7444$, Equation (10.4) gives

$$(\widetilde{Q}_{0.10}, \widetilde{Q}_{0.50}, \widetilde{Q}_{0.90}) = (-0.2741, 0.2000, 0.9444). \tag{10.9}$$

If the frozen target scaler for that cell has $\mu^Y = 0$ and $s^Y = 4$ ticks, the unnormalized offsets are

$$(-1.0964, 0.8000, 3.7776) \text{ ticks}. \tag{10.10}$$

With $p_{\mathrm{ref}} = 100$ and $\delta_p = 0.5$, the pre-snapping physical POCs are $(99.4518, 100.4000, 101.8888)$. This is an illustrative arithmetic trace, not a reported model prediction. If the same horizon's presence logits were $(2, -1)$, their probabilities would be $(\mathrm{sigmoid}(2), \mathrm{sigmoid}(-1)) = (0.8808, 0.2689)$.

11 Training Objective

11.1 Pinball loss and its quantile condition

For level $\tau \in (0, 1)$, outcome y, prediction q, and residual $u = y - q$, the pinball loss is

$$\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\}) = \max\{\tau u, (\tau - 1)u\}. \tag{11.1}$$

Why the minimizer is a conditional quantile

For conditional distribution $F_{Y|X}$, let $R(q \mid X) = \mathbb{E}[\rho_\tau(Y - q) \mid X]$. Away from atoms,

$$\begin{aligned} \frac{\partial R}{\partial q} &= (1 - \tau)\Pr(Y < q \mid X) - \tau \Pr(Y > q \mid X) \\ &= F_{Y|X}(q) - \tau. \end{aligned} \tag{11.2}$$

A stationary minimum therefore satisfies $F_{Y|X}(q) = \tau$. At an atom, the subgradient condition is $F(q^-) \le \tau \le F(q)$, which is the standard quantile definition. The property pertains to the target defined in Equation (2.4); it does not change that target's semantics.

Let horizon weights $w_h = h^{-1/2}$, target weights $v = (1, 1, 1, 0.5, 0.5)$, quantile levels $\mathcal{T} = \{0.1, 0.5, 0.9\}$, and target mask $m^Y_{b,h,k}$. The normalized primary loss is

$$\mathcal{L}_Q = \frac{\sum_{b,h,k,\tau} m^Y_{b,h,k} w_h v_k \rho_\tau(\widetilde{y}_{b,h,k} - Q^\star_{b,h,k,\tau})}{\sum_{b,h,k,\tau} m^Y_{b,h,k} w_h v_k + \epsilon}. \tag{11.3}$$

11.2 Median stabilization and presence loss

For normalized median residual r and threshold $\delta = 1$,

$$H_\delta(r) = \begin{cases} \frac{1}{2} r^2, & |r| \le \delta, \\ \delta(|r| - \frac{1}{2}\delta), & |r| > \delta. \end{cases} \tag{11.4}$$

The masked, weighted median loss is

$$\mathcal{L}_H = \frac{\sum_{b,h,k} m^Y_{b,h,k} w_h v_k H_1(\widetilde{y}_{b,h,k} - Q^\star_{b,h,k,0.5})}{\sum_{b,h,k} m^Y_{b,h,k} w_h v_k + \epsilon}. \tag{11.5}$$

For presence label $z \in \{0, 1\}$ and logit η, stable binary cross-entropy is

$$\mathrm{BCE}(\eta, z) = \max(\eta, 0) - \eta z + \log(1 + e^{-|\eta|}). \tag{11.6}$$

With an independent presence-validity mask m^P, define

$$\mathcal{L}_P = \frac{\sum_{b,h,s} m^P_{b,h,s}\,\mathrm{BCE}(\eta_{b,h,s}, z_{b,h,s})}{\sum_{b,h,s} m^P_{b,h,s} + \epsilon}. \tag{11.7}$$

The same shared head applied after decoder block one supplies an auxiliary deep-supervision loss. For optimizer-step fraction $u \in [0,1]$,

$$\lambda_{\mathrm{aux}}(u) = \begin{cases} 0.25, & u \le 0.60, \\ 0.25 - \dfrac{0.20}{0.30}(u - 0.60), & 0.60 < u < 0.90, \\ 0.05, & u \ge 0.90. \end{cases} \tag{11.8}$$

With $\mathcal{L}_{\mathrm{base}} = \mathcal{L}_Q + 0.25\mathcal{L}_H + 0.20\mathcal{L}_P$, the total objective is

$$\mathcal{L} = \mathcal{L}^{\mathrm{final}}_{\mathrm{base}} + \lambda_{\mathrm{aux}}(u)\mathcal{L}^{\mathrm{block}\ 1}_{\mathrm{base}}. \tag{11.9}$$

12 Parameter and Attention-Work Derivation

12.1 Primitive formulas

A biased linear map $a \to b$ contains

$$P_{\mathrm{linear}}(a,b) = ab + b \tag{12.1}$$

parameters. A SwiGLU map $d \to f \to d$ has two input projections and one output projection:

$$P_{\mathrm{SwiGLU}}(d,f) = 2(df + f) + (fd + d) = 3df + 2f + d. \tag{12.2}$$

For a main temporal CHMT block with $d = 128$, $f = 256$, and four heads,

$$\begin{aligned} P_{\mathrm{attn}} &= 4(d^2 + d) = 66{,}048, \\ P_{\mathrm{SwiGLU}} &= 3df + 2f + d = 98{,}944, \\ P_{\mathrm{norm+scale}} &= 2d + 2d = 512, \\ P_{\mathrm{temperature+time}} &= 4 + 4 = 8, \\ P_{\mathrm{temporal\ block}} &= 165{,}512. \end{aligned} \tag{12.3}$$

There are nine such blocks, so their total is $9 \times 165{,}512 = 1{,}489{,}608$.

For the L2 level block, $d = 32$, $f = 64$, and four heads. It has

$$4(32^2 + 32) + (3 \cdot 32 \cdot 64 + 2 \cdot 64 + 32) + 4 \cdot 32 + 4 = 10{,}660, \tag{12.4}$$

where the final four scalars are head temperatures; level attention has no elapsed-time decay.

One decoder block contains self-attention, cross-attention, one SwiGLU, three RMSNorm gains, three LayerScale vectors, and eight head temperatures:

$$2 \cdot 66{,}048 + 98{,}944 + 3 \cdot 128 + 3 \cdot 128 + 8 = 231{,}816. \tag{12.5}$$

Two blocks therefore contain 463,632 parameters.

12.2 Literal component ledger

Component	Trainable parameters
Bar, L1, and L2-level projections	4,768
Quality MLP	18,688
Bar time MLP	4,768
L1 patch-time MLP	4,832
L2 patch-time MLP	4,832
Three modality embeddings	384
L2 side and rank embeddings	384
L2 level CHMT block	10,660
Book query, pool norm, and 32-to-128 projection	4,288
L1/L2 within-patch embeddings and pool MLPs	9,346
Nine temporal CHMT blocks	1,489,608
Two decoder blocks	463,632
Three modality-gate MLPs	49,539
Fusion norm and context projection	16,640
Horizon queries and horizon MLP	4,992
Shared 128-to-17 output head	2,193
Total	**2,089,554**

Table 8: Exact trainable-parameter ledger. The total includes 36 learned per-head Q/K temperatures in the nine temporal blocks.

Several rows can be checked directly. The quality MLP is

$$P_q = (16 \cdot 128 + 128) + (128^2 + 128) = 18{,}688. \tag{12.6}$$

Each modality gate is $256 \to 64 \to 1$:

$$P_{\text{gate}} = (256 \cdot 64 + 64) + (64 \cdot 1 + 1) = 16{,}513, \tag{12.7}$$

and $3P_{\text{gate}} = 49{,}539$. The shared head contains $128 \cdot 17 + 17 = 2{,}193$ parameters. The two patch modules each contain four 128-vectors plus a $128 \to 32 \to 1$ scoring MLP:

$$P_{\text{patch}} = 4 \cdot 128 + (128 \cdot 32 + 32) + (32 + 1) = 4{,}673, \tag{12.8}$$

giving 9,346 across L1 and L2.

12.3 Why dense attention is small here

Per layer and head, all temporal score matrices contain

$$64^2 + 60^2 + 60^2 = 11{,}296 \tag{12.9}$$

entries. Across three layers and four heads, that is

$$3 \cdot 4 \cdot 11{,}296 = 135{,}552. \tag{12.10}$$

Two decoder cross-attention layers add

$$2 \cdot 4 \cdot 5 \cdot 184 = 7{,}360 \tag{12.11}$$

entries. For one batch element, the single level block across all 240 snapshots contributes

$$240 \cdot 4 \cdot 20^2 = 384{,}000 \tag{12.12}$$

entries, and two decoder self-attention layers contribute

$$2 \cdot 4 \cdot 5^2 = 200. \tag{12.13}$$

The complete multi-head Q/K attention-score count for one forward pass is therefore

$$135{,}552 + 384{,}000 + 7{,}360 + 200 = 527{,}112. \tag{12.14}$$

Book-query pooling evaluates another $240 \cdot 20 = 4{,}800$ scalar compatibility scores, and the two patch pools evaluate $2 \cdot 60 \cdot 4 = 480$ MLP scores; neither is a multi-head Q/K matrix. At these sequence lengths, dense temporal attention score storage is modest; the repeated projections, feed-forward layers, and 240 snapshot-level passes dominate arithmetic.

13 Reference Forward Pass and Inference Contract

The complete computation is the following ordered map:

1. validate all external shapes, dtypes, finite values, and the nested L2 mask invariant in Equation (3.3);
2. robust-scale valid numeric inputs with frozen training statistics and zero invalid rows;
3. project bars and L1, and project each L2 side–rank cell with internal side/rank embeddings;
4. run one level CHMT block and book-query pool for every L2 snapshot;
5. pool each nonoverlapping group of four L1 and L2 frames; do not patch bars;
6. add modality and physical-time embeddings, then run three causal CHMT blocks independently on each branch;
7. extract latest-valid summaries, compute quality context and the three hard-aware gates, and concatenate the 184 memory tokens;
8. construct five horizon queries, then apply two decoder blocks in the self-attention → cross-attention → SwiGLU order;
9. map the five final states to $[B, 5, 17]$ raw values;
10. construct ordered quantiles and presence probabilities, reverse target normalization, reconstruct physical values, snap prices, and apply all output invariants.

The raw graph returns only R from Equation (10.1). A serving wrapper binds the model bytes to feature-schema, scaler, target-transform, calibration, tick-rule, and model-version hashes. A valid inference result must satisfy

$$\begin{aligned} &R \in \mathbb{R}^{1\times 5\times 17}, \qquad Q_{0.10} \le Q_{0.50} \le Q_{0.90}, \\ &\widehat{V} \ge 0, \qquad -1 \le \widehat{f}_D \le 1, \qquad 0 \le \widehat{p}^{\,\text{side}} \le 1. \end{aligned} \tag{13.1}$$

Any schema mismatch, stale reference, all-modality mask, nonfinite result, or invariant violation yields a reason-coded abstention rather than a partial forecast.

14 Related Design Choices

CHMT-Q uses dense attention because its temporal sequences contain only 64 or 60 tokens. Relative elapsed-time bias adapts relative-position attention [2] to physical clock differences. Patch pooling is related to PatchTST's use of temporal patches [5], but CHMT-Q jointly embeds interacting numeric fields rather than enforcing channel independence. Direct horizon queries resemble multi-horizon decoding in Temporal Fusion Transformer [6]; the hard masks and explicit quality routes are specific to the present contract.

The depth hierarchy follows the broad lesson of DeepLOB and later LOB models: same-snapshot spatial structure and across-time evolution deserve separate processing [7–9]. Unlike a flattened book, CHMT-Q lets rank and side interact before reducing one snapshot to one temporal frame. Its ordered head uses quantile regression, whose proper-score interpretation is established by the pinball-loss result [12]. The marginals could later be calibrated on a dedicated calibration partition, for example by conformalized quantile methods under their stated assumptions [13]; calibration is not implicit in the architectural ordering guarantee.

15 Implementation Status and Research Boundary

The tensor shapes, primitive blocks, and 2,089,554-parameter count in this paper correspond to the implemented CHMT-Q model and its frozen architecture ledger. The model has been trained in an offline research study. It has no established live-terminal integration or production-deployment status. This mechanics paper reports neither an independent performance estimate nor an economic result, and it makes no claim of executable edge or profitability. Existing evidence does not establish useful conditional medians, executable edge, or profitability. The artifact remains research-only and non-promotable.

The model is a forecast provider. It does not choose side, position size, leverage, stop, venue, or order type. Any downstream consumer must preserve the cutoff and scaler contracts, distinguish a forecast from realized footprint data, and evaluate decision utility, costs, risk, and operational behavior in its own protocol.

Raw exchange archives, compiled tensors, prediction tapes, model weights, and resumable checkpoints are not included in the paper source package. Their distribution requires a separate provenance, licensing, privacy, and release decision. The source bundle contains the manuscript and the figures necessary to reproduce this document; it is not a public end-to-end reproduction package.

16 Conclusion

CHMT-Q maps four masked numeric market views to five direct probabilistic forecasts through a structure-preserving Transformer. Its defining operations are explicit: 64 unpatched bar tokens; 60 pooled L1 tokens; 60 hierarchical L2 tokens; a 184-token gated memory; five horizon queries; and a 17-value raw head at each horizon. The output map is equally explicit: five center/gap triplets become ordered q10/q50/q90 marginals, two logits become side-POC presence probabilities, and frozen inverse transforms recover physical price, volume, and delta-fraction units. Every step retains an authoritative validity mask.

The detailed derivations make the model auditable at the level where numeric market systems most often become ambiguous: which observation is legal at the cutoff, what every axis means, which values can attend, what each raw output slot encodes, and which transformation returns it to physical units. Those contracts define CHMT-Q more precisely than the word "Transformer" alone.

A Compact Symbol Table

Table 9: Principal notation.

Symbol	Meaning
B	batch size; fixed to one for exported inference
t_c	wall-clock feature boundary and first target-interval opening
$c = t_c^-$	logical total-order cutoff immediately before the target-open phase
κ_i	deterministic total-order key of captured record i
$\mathcal{F}_c$	information available by cutoff
h, H	horizon index and horizon count; $H = 5$
k	target index; total POC, buy POC, sell POC, log-volume residual, delta fraction
p_{ref}	fresh causal reference price
δ_p	instrument tick size
ℓ_{ref}	causal trailing log-volume reference
d, d_h, d_{ff}	model, attention-head, and feed-forward widths
m	Boolean validity or attention-eligibility mask
α_m	normalized quality gate for modality m
z_{ctx}	fused 128-dimensional quality and modality context
$r_{h,k}$	raw center parameter for horizon h, target k
Q_τ	conditional quantile at level τ
ρ_τ	pinball loss at level τ
θ	all trainable model parameters

B Mask Algebra

For a temporal query i and key ℓ, the encoder eligibility indicator is

$$e_{i\ell} = m_i^{\text{query}} m_\ell^{\text{key}} \mathbb{I}\{t_\ell \le t_i\}. \tag{B.1}$$

For L2 level attention inside snapshot t,

$$e_{(s,r),(s',r')} = m_{t,s,r}^{2,L} m_{t,s',r'}^{2,L}, \tag{B.2}$$

because all cells share one timestamp. For decoder cross-attention,

$$e_{h,\ell} = m_{\ell}^{\text{memory}}, \tag{B.3}$$

and the learned quality gate appears only as a finite score bias. These three relations separate temporal causality, structural coexistence, and reliability routing.

C Source-Package Manifest

The submission archive contains this LaTeX source and the three referenced PNG figures. It intentionally excludes raw or compiled market data, prediction tapes, weights, checkpoints, credentials, and licensed feeds. The document is independently compilable with the figure directories preserved.

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